

activations and UGS are competitive with EAP-IG-inputs w.r.t. CMD, but have higher runtime and memory usage respectively. For CPR, they are less competitive; this is unsurprising for UGS, as it directly optimizes for maintaining model behavior, rather than for finding solely positively impactful components.

More surprisingly, edge activation patching (EActP) does not always perform best, despite computing *exact* IEs for each edge: it dominates for IOI on GPT-2, but not Qwen-2.5 or InterpBench. This could stem from our use of fewer examples to run EActP due to its long runtime. But EActP also has deeper limitations: like attribution methods, it estimates the effect of ablating each edge independently; this may imperfectly predict the effect of ablating multiple edges in tandem (Mueller, 2024). That said, UGS, which considers many edges at once, is also not the top performer.

Circuits found with CF ablations outperform those found with mean or optimal ablations; the latter two score similarly to each other. This is expected, as the setting in which CF circuits are found more closely resembles the evaluation setting, and more precisely localizes the distinction captured by the CF input pairs.

Certain methods underperform with respect to both metrics: node-level circuits do poorly, likely because each node included “costs” many edges; such circuits simply cannot be as sparse as edge circuits. The non-causal method IFR tends to achieve lower performance than attribution methods, but significantly better than random circuits in most cases.

In summary, **EAP-IG-inputs achieves the highest performance** on average on both CMD and CPR. However, **other techniques**, like EAP-IG-activations, EAP, and UGS, **remain competitive**.

4. Causal Variable Localization Track

The circuit localization track evaluates methods that localize *behaviors* to model components that form *end-to-end pathways* from input to output. In contrast, the causal variable localization track evaluates methods that localize a specific *concept* along causally active paths. Figure 3 visually depicts an example evaluation in this track, while Table 3 contains results for causal variable localization methods.

4.1. Causal Abstraction

A basic assumption in mechanistic interpretability is that models implement intelligent behaviors by representing and manipulating concepts.¹¹ We operationalize such hypotheses by encoding the reasoning process as a causal model $\mathcal{H}$ with variables corresponding to concepts. The task is to

¹¹Crucially, the concepts employed by an LM may not relate to those employed by a human on the same task (Hewitt et al., 2025).

align these high-level conceptual variables in a causal model with low-level features in a neural model that have the same mechanistic role, i.e., the high-level causal model is a *causal abstraction* of the LM (Geiger et al., 2021; 2024a).

LM Features. What should be the atomic units of analysis, or *features*, for mechanistic interpretability? The answer is hotly debated and not currently clear (Mueller et al., 2024), so we design this track to incentivize investigation of this fundamental question. We adopt the framework of Geiger et al. (2024a) in which any hidden vector $\mathbf{h} \in \mathbb{R}^d$ constructed by a model $\mathcal{N}$ during inference can be mapped into a new feature space $\mathbb{F}^k$ (e.g., a rotated vector space) using an invertible function $\mathcal{F} : \mathbb{R}^d \rightarrow \mathbb{F}^k$ (e.g., multiplication with an orthogonal matrix). Features Π are a set of indices between 1 and k , i.e., a set of dimensions in $\mathbb{F}^k$. This framework supports a variety of features, including neurons, orthogonal directions, SAE features, and non-linear features. The vector $\mathbf{h}$ might come from the residual stream between transformer layers or the output of an attention head.

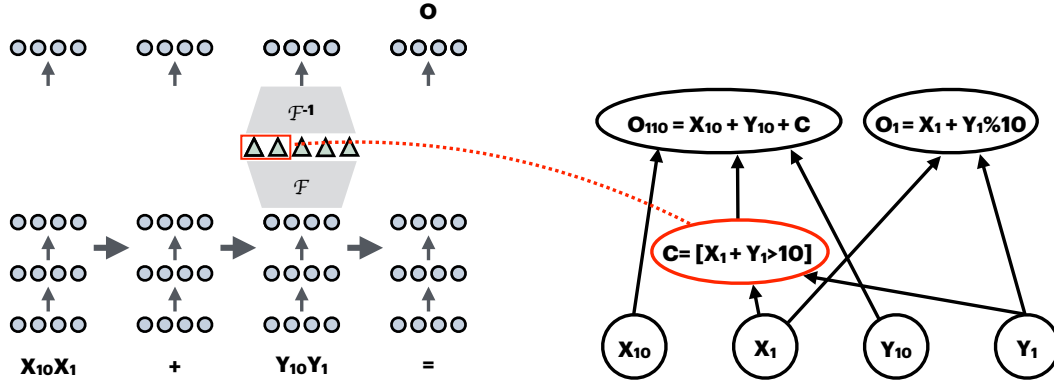
Alignments. Alignments between high-level conceptual variables and low-level features will not be static, even in the simplest cases. For instance, in MCQA, the index of the token corresponding to the correct multiple choice answer will change depending on the number of tokens in the question. As such, submissions to MIB can provide an **alignment** that aligns a variable X with features Π_X of a dynamically selected hidden vector $\mathbf{h}$, e.g., the residual stream of the correct answer token in the MCQA task.

Faithfulness metrics. We quantify the degree to which a variable $X \in \mathcal{H}$ faithfully abstracts the features Π_X using *interchange interventions*. Given base and counterfactual inputs (b, c) , the interchange intervention $\mathcal{H}_{X \leftarrow \text{Get}(\mathcal{H}(c), X)}(b)$ runs $\mathcal{H}$ on base input b while fixing the variable X to the value it takes when $\mathcal{H}$ is run on a counterfactual input c (Vig et al., 2020; Geiger et al., 2020). The distributed interchange intervention $\mathcal{N}_{\Pi_X \leftarrow \text{Get}(\mathcal{N}(c), \Pi_X)}(b)$ runs $\mathcal{N}$ on b while fixing the features Π_X of the hidden vector $\mathbf{h}$ passed through $\mathcal{F}$ to the value they take for counterfactual input c (Wu et al., 2023; Amini et al., 2023; Geiger et al., 2024b).

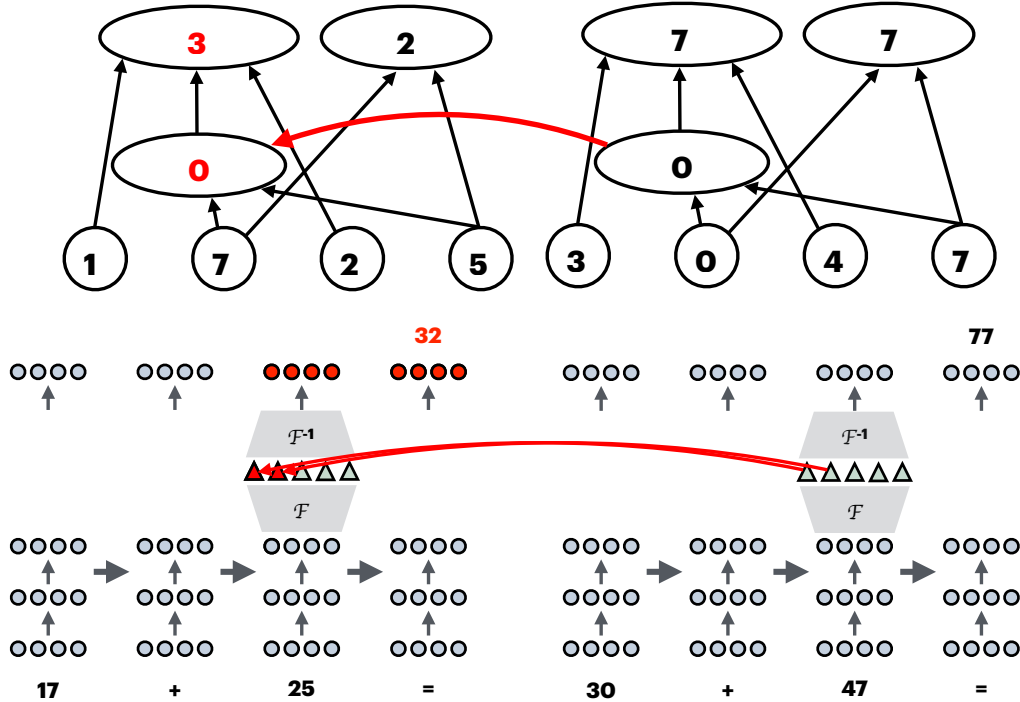
Given a counterfactual dataset $\mathcal{D}$ and a high-level causal model $\mathcal{H}$ aligned to an LM $\mathcal{N}$, we measure whether interchange interventions on a variable X in $\mathcal{H}$ and aligned features Π_X in $\mathcal{N}$ produce the same output:

$$\text{Faith}(X, \Pi_X, \mathcal{H}, \mathcal{D}) = \sum_{(b, c) \in \mathcal{D}} [\mathcal{H}_{X \leftarrow \text{Get}(\mathcal{H}(c), X)}(b) = \mathcal{N}_{\Pi_X \leftarrow \text{Get}(\mathcal{N}(c), \Pi_X)}(b)].$$

This metric of *interchange intervention accuracy* (IIA) is for all tasks except IOI, which has a logit-based metric (§4.6).



(a) **Arithmetic Task Submission.** Users submit an alignment between the carry-the-one variable X_{Carry} in a high-level causal model $\mathcal{H}_+$ and two features $\Pi_{X_{\text{Carry}}}$ of the neural network’s residual stream at the second number token.



(b) **Arithmetic Task Evaluation.** An aligned interchange intervention with base input “17+25=” and counterfactual input “30+47=”. At the high-level, the interchange intervention $\mathcal{H}_{X_{\text{Carry}} \leftarrow \text{Get}(\mathcal{H}(30+47), X_{\text{Carry}})}(17+25)$ fixes the carry-the-one variable X_{Carry} to the value 0 (from the counterfactual) instead of its natural value 1 (from the base input), causing the causal model to output “32” instead of “42”. At the low-level, the interchange intervention $\mathcal{N}_{\Pi_{X_{\text{Carry}}} \leftarrow \text{Get}(\mathcal{N}(30+47), \Pi_{X_{\text{Carry}}})}(17+25)$ fixes the aligned features $\Pi_{X_{\text{Carry}}}$ of the LM to the value they take on when the LM is run on the counterfactual input. The low-level output after intervention “32” is equal to the high-level output after intervention, which is a piece of evidence supporting the hypothesized alignment between the carry-the-one variable and the identified neural network features. The faithfulness metrics aggregate these individual experiments across base-counterfactual input pairs.

Figure 3. A schematic of the causal variable localization track submission and evaluation. Users submit an alignment between a high-level causal variable X and hidden vector features Π_X in an LM (top). In evaluations, aligned interchange interventions are performed with base and counterfactual inputs (b, c) on the high-level causal model $\mathcal{H}_{X \leftarrow \text{Get}(\mathcal{H}(c), X)}(b)$ and the low-level neural network $\mathcal{N}_{\Pi_X \leftarrow \text{Get}(\mathcal{N}(c), \Pi_X)}(b)$. The more similar LM output under intervention is to the causal model output under intervention, the more faithfully the causal model abstracts the LM (bottom).

For each task, we have base and counterfactual input pairs (defined in §2.1). We use interchange interventions on counterfactual pairs to isolate variables in $\mathcal{H}$ that result in different outputs, conditional on which variables are targeted in the intervention. This is not as obvious as it first appears; naïve approaches to sampling counterfactual inputs used for intervention can undersample or exclude crucial settings.

For all tasks, we filter out all examples where the model predicts the incorrect output for the base input or any of the counterfactuals used. Possible extensions may expand to explain when LMs fail, which would be analogous to the CMD metric from the circuit localization track.

4.2. Causal Variable Localization Baselines

Featurizers and feature selection. We consider five baselines for constructing the featurizer $\mathcal{F}$ and selecting the features Π_X . See App. F for further details.

We evaluate three *unsupervised* featurization methods that provide features without access to a high-level causal model. The most naïve of these is the “Full Vector” baseline; this entails using an identity featurizer, and then selecting all features—i.e., intervening on the full untransformed hidden vector $\mathbf{h}$. We also evaluate PRINCIPAL COMPONENT ANALYSIS (PCA; Tigges et al. 2023; Marks & Tegmark 2024) and SPARSE AUTOENCODERS (SAE; Bricken et al. 2023; Huben et al. 2024), which encode into very high-dimensional spaces with many features. For SAEs, we use the publicly available GemmaScope (Lieberum et al., 2024) and LlamaScope (He et al., 2024).

To select SAE features, principal components, or standard dimensions of hidden vectors that are aligned with high-level causal variables, we use DESIDERATA-BASED MASKING (DBM; Cao et al. 2020; 2022; Csordás et al. 2021; Davies et al. 2023; Chaudhary & Geiger 2024) to learn a binary mask over features using a high-level causal model as a source of supervision. The masks are trained to maximize the faithfulness metric on training data.

We also evaluate a *supervised* featurization method DISTRIBUTED ALIGNMENT SEARCH (DAS; Geiger et al. 2024b) that learns a featurizer with supervision from the high-level causal model. First, a variable in the high-level causal model is aligned with features that are randomly initialized orthogonal directions that define a linear subspace of a hidden vector. Then, the features are trained to maximize the faithfulness metric. There is no need for a separate feature selection procedure because the features are constructed specifically for the high-level variable.

Dynamic alignment. For tasks other than IOI, we brute-force search over a few manually selected token locations. For each layer and token location, we attempt to locate

features of the residual stream vector that correspond to variables in the high-level causal model. For each causal variable, we use training data to create and select features, and then evaluate the faithfulness of aligning those features with the causal variable. We compute the token location with the highest score at each layer and report the best layer and the average across layers. Future submissions are free to target any tokens. For IOI, we focus our baseline experiments to the attention heads identified by Wang et al. (2023) in GPT2-small, but we will allow for future submissions to identify new attention heads in the other three models.

4.3. MCQA and ARC (Easy)

Causal model. For the two multiple-choice question answering datasets, we hypothesize the LM computes the position of the answer token in context before retrieving the answer token itself. The high-level causal model $\mathcal{H}_{\text{MCQA}}$ is a simple algorithm with three variables: a text input variable T , an ordering ID X_{Order} (Dai et al., 2024) storing the position of the answer, and the answer token O_{Answer} . This model abstracts away the details of how the answer position is computed; the mechanism for X_{Order} is a lookup table from inputs to the index of the answer token, i.e., a number between 1 and 4 because there are four choices. Instead, the focus is on the retrieval of the correct choice; the mechanism for O_{Answer} dereferences the index stored in X_{Order} .

Counterfactuals. We use counterfactuals where the answer position is changed, where the choice letters are randomized, or both. When both the answer position and the choice letters are different in the counterfactual, a different output is expected when localizing the ordering id X_{Order} versus the answer token O_{Answer} . If the ordering ID is targeted, then the expected output is the choice token in the base at the answer position from the counterfactual. If the answer token is targeted, then the expected output is the answer token from the counterfactual, regardless of position.

Results. We target the residual stream of the last token of the input and the correct choice letter token at each layer. We generally see strong evidence (Tables 3a and 3c; App. F.5.1) of the causal model $\mathcal{H}_{\text{MCQA}}$ being a faithful abstraction, with DAS successfully disentangling the ordering ID variable X_{Order} from the output token O_{Answer} in many layers. Even the full vector baseline successfully localizes the variable in some layers, though it performs poorly on average because both variables are aligned with the same features.

4.4. Two-Digit Addition

Causal model. For the two-digit addition task, we hypothesize that LMs use a “carry-the-one” algorithm, as illustrated in Figure 3. The causal model $\mathcal{H}_+$ has a text input variable T that is parsed into the variables X_1, X_{10} ,